IONIC EQUILIBRIUM

AKK SIR (AKHILESH KANTHER)

Q. 100ml, 0.1 M HCl is mixed 0.1 M NaOH

Volume of NaOH added	$[H^+]$ or $[OH^-]$	pH
0 ml	$[H^+] = 0.1M$	1 B 11 − 13 R
90 ml	$[H^+] = \frac{1}{190} = \frac{1}{1.9} \times 10^{-2}$	$2 - \log\frac{1}{1.9} = 2.3$
99 ml	$[H^+] = \frac{0.1}{199} = \frac{0.1}{200} = \frac{1}{2} \times 10^{-3}$	$3 - \log 1/2 = 3.3$
100ml		B 5 − 7 R
101 ml	$[OH^-] = \frac{0.1}{201} = \frac{0.1}{200} = \frac{1}{2} \times 10^{-3}$	pOH = 3.3, pH = 10.7

Q. 100ml, 0.1 M HCl is mixed 0.1 M NaOH

Volume of NaOH added	$[H^+]$ or $[OH^-]$	pH
0 ml	$[H^+] = 0.1M$	1
90 ml	$[H^+] = \frac{1}{190} = \frac{1}{1.9} \times 10^{-2}$	$2 - \log\frac{1}{1.9} = 2.3$
99 ml	$[H^+] = \frac{0.1}{199} = \frac{0.1}{200}$ $= \frac{1}{2} \times 10^{-3}$	$3 - \log 1/2 = 3.3$
101 ml	$[OH^-] = \frac{0.1}{201} = \frac{0.1}{200}$	$pOH = 3.3, pH = 10.7$

In acid base titration, pH changes suddenly near the equivalence point. This sudden change in pH depends on acid-base pair being titrated.

The selection of indicator must satisfy the condition that "pH range of indicator must lie within this sudden jump or fall".

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$SA \rightarrow 3$

$SB \rightarrow 11$

$WA \rightarrow 6$

$WB \rightarrow 8$

(1) Titration of Strong acid with strong Base

e.g. $HCl + NaOH$

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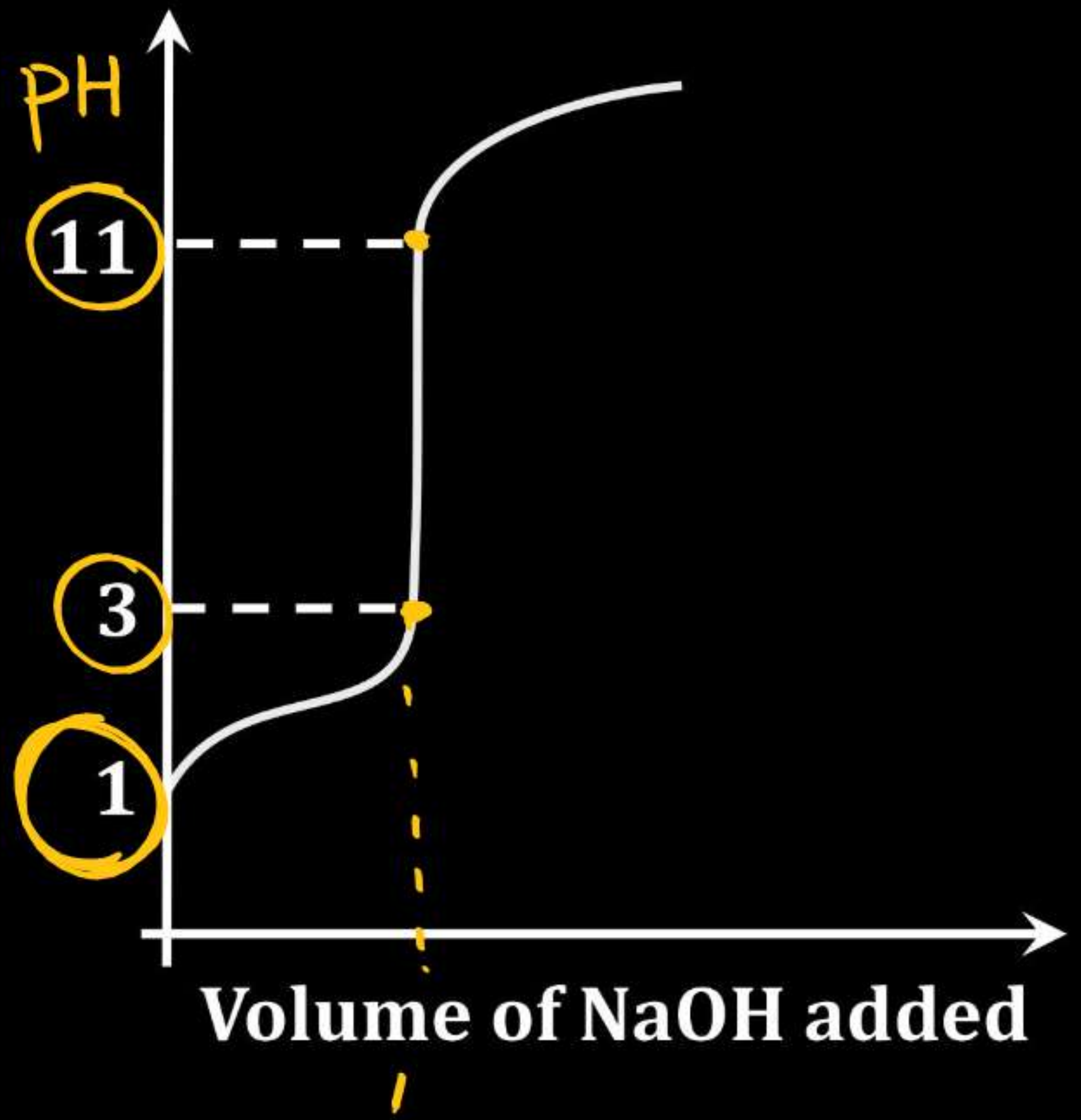
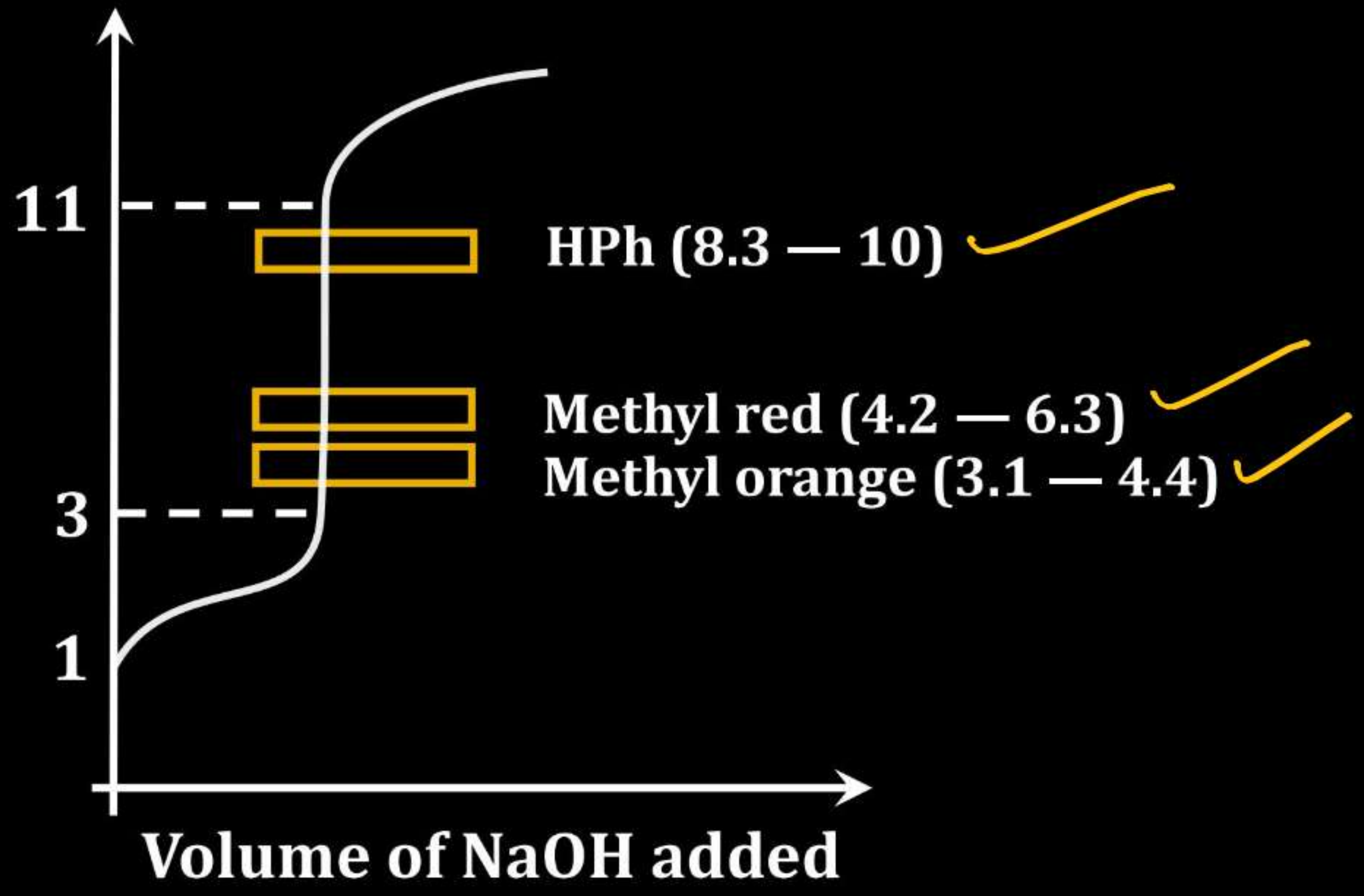
e.g. HCl + NaOH



(1) Titration of Strong acid with strong Base

e.g. HCl + NaOH

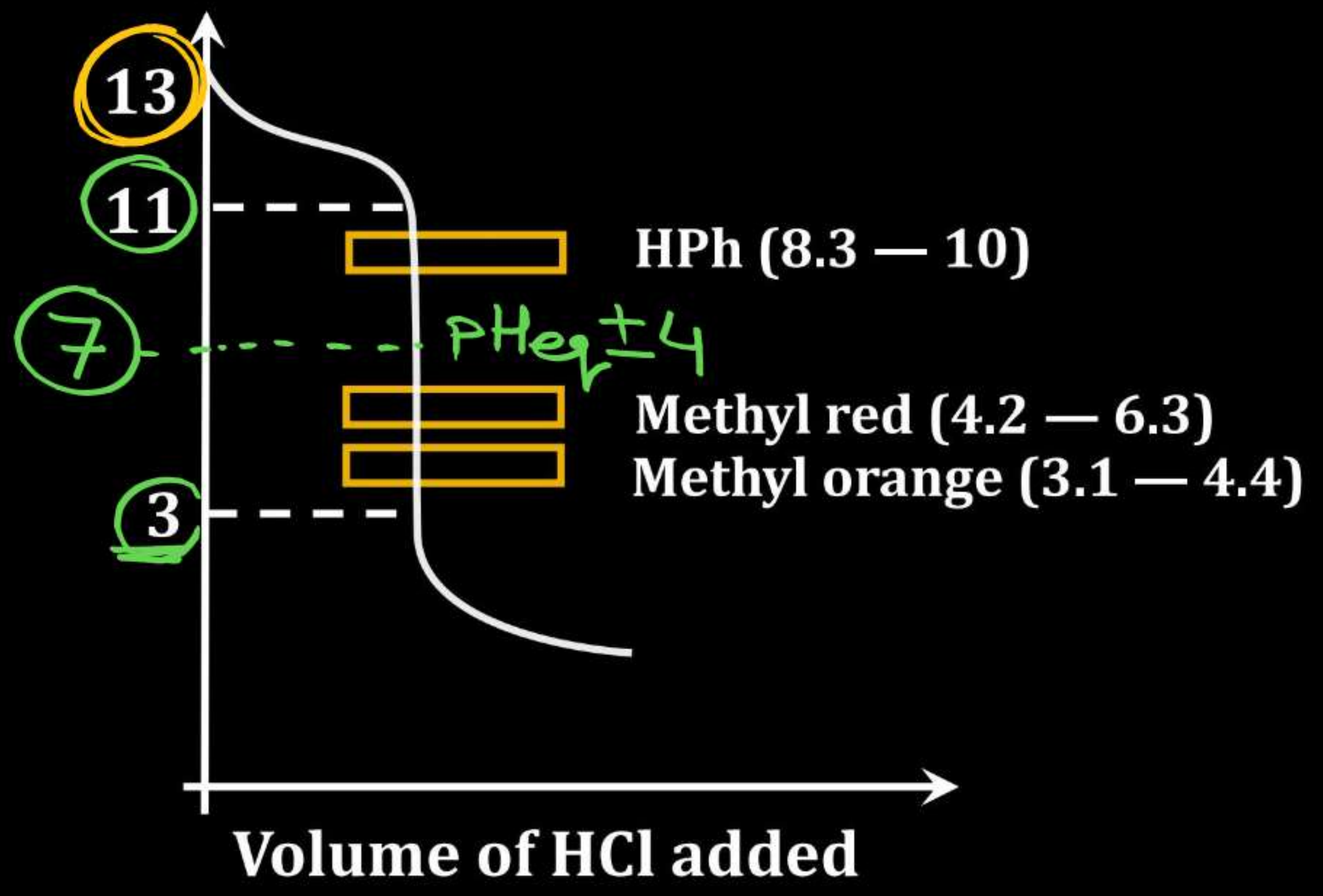


(2) **Titration of Strong base with strong acid** :

e.g. $NaOH + HCl$

(2) Titration of Strong base with strong acid

e.g. $NaOH + HCl \rightarrow NaCl$



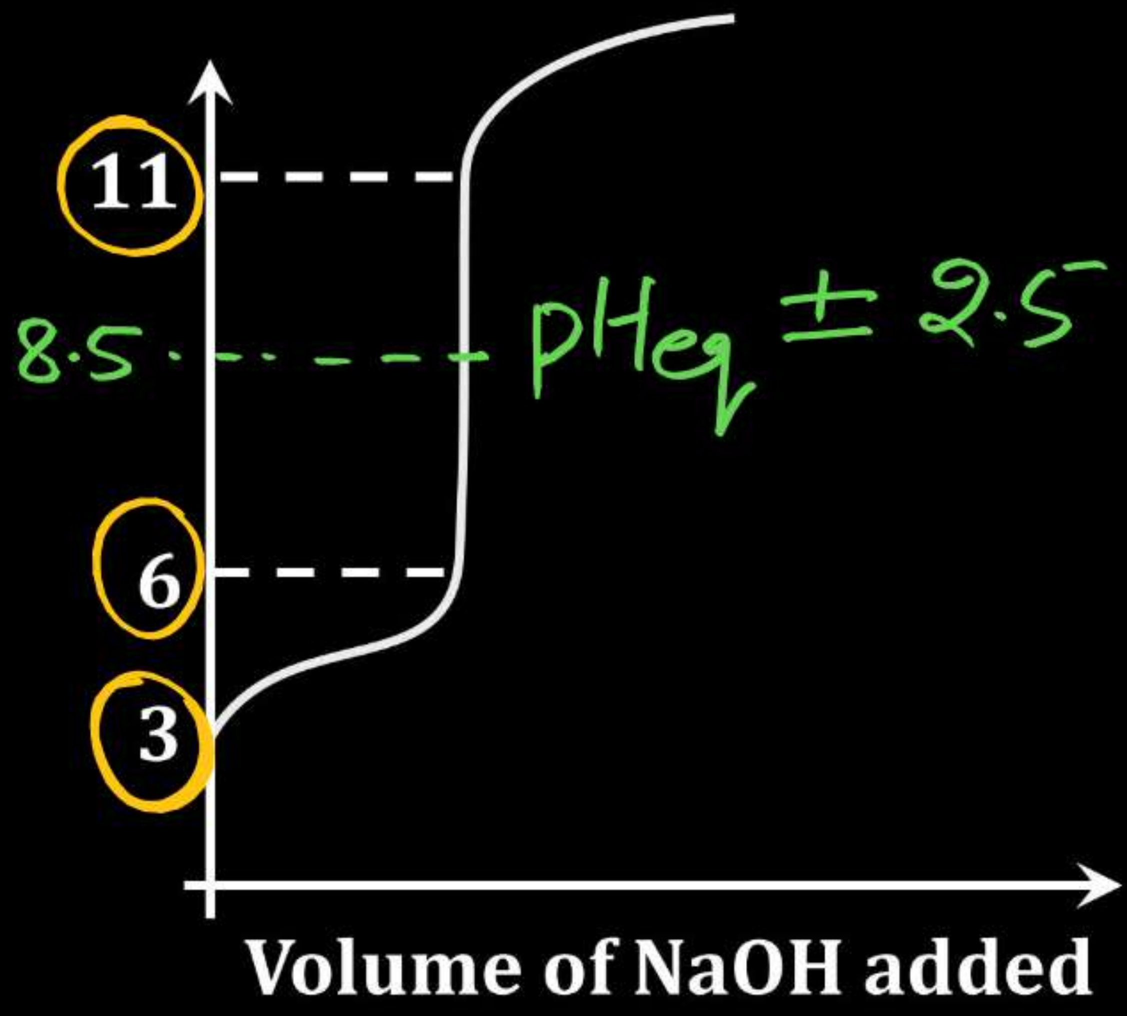
(3) Titration of weak acid with strong Base

e.g. $CH_3COOH + NaOH$

CH_3COONa

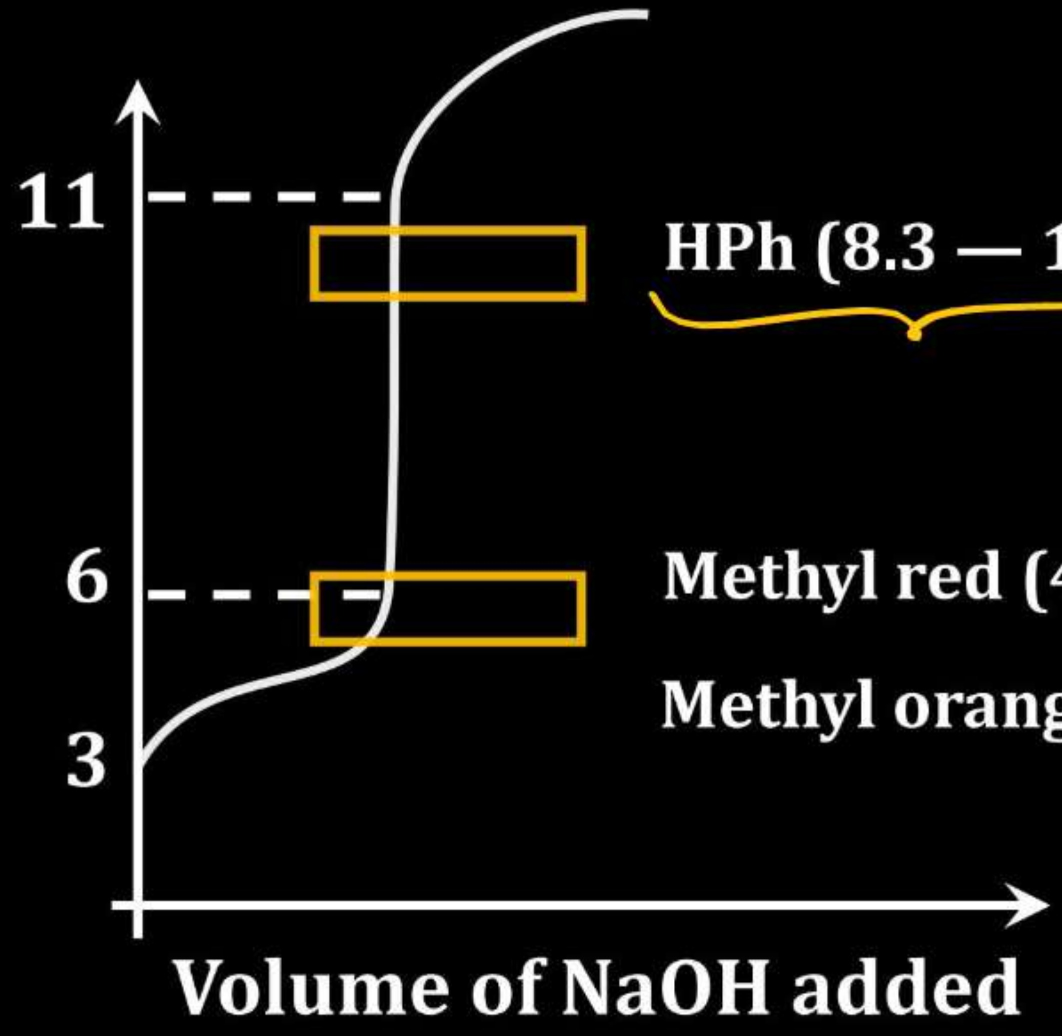
(3) Titration of weak acid with strong Base

e.g. $CH_3COOH + NaOH \longrightarrow CH_3COONa$



(3) Titration of weak acid with strong Base

e.g. $CH_3COOH + NaOH$



HPh (8.3 — 10)

Methyl red (4.2 — 6.3) → not suitable

Methyl orange (3.1 — 4.4) → not suitable

(4) Titration of weak base with strong acid

e.g. $NH_4OH + HCl$

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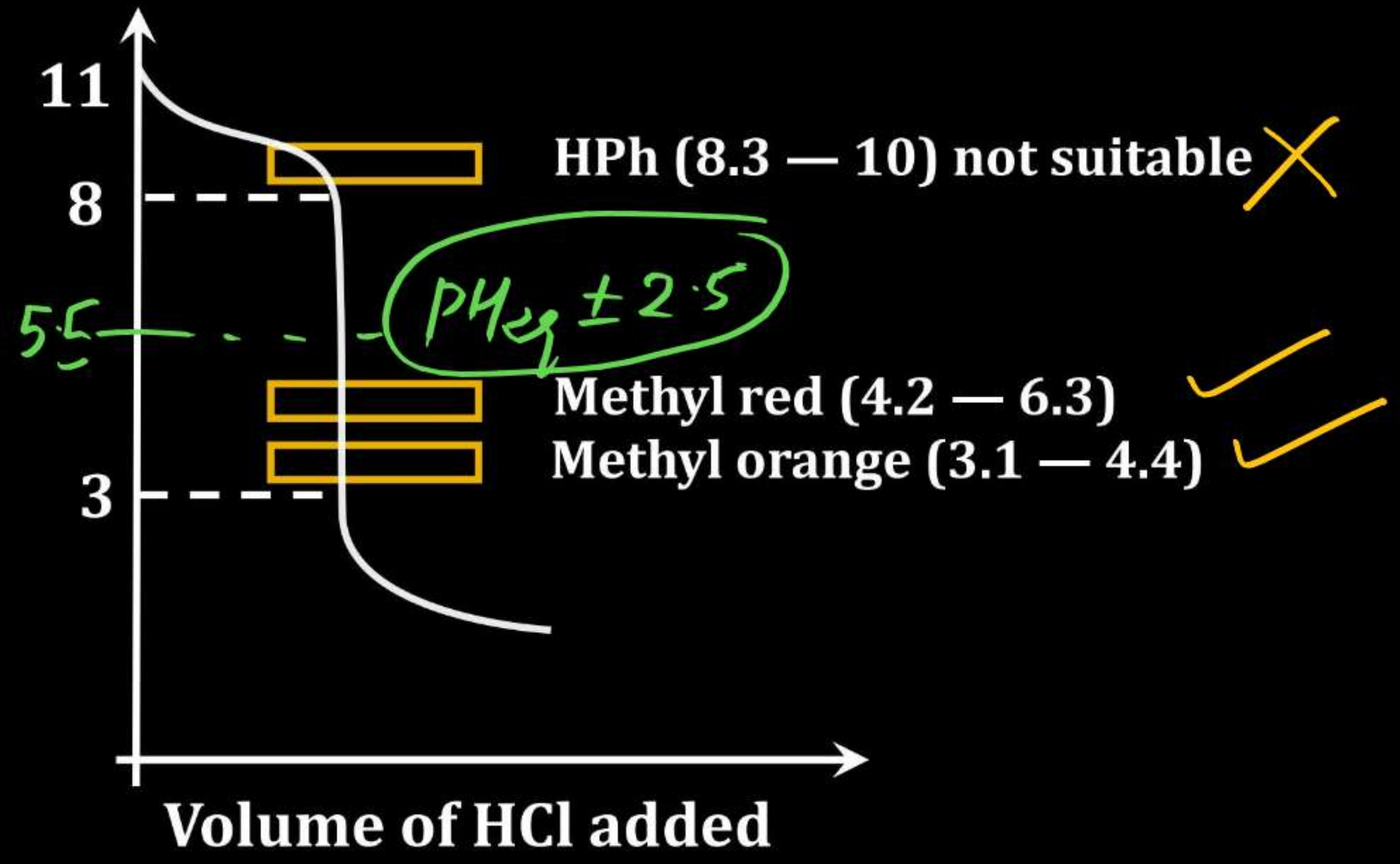
e.g. $NH_4OH + HCl$



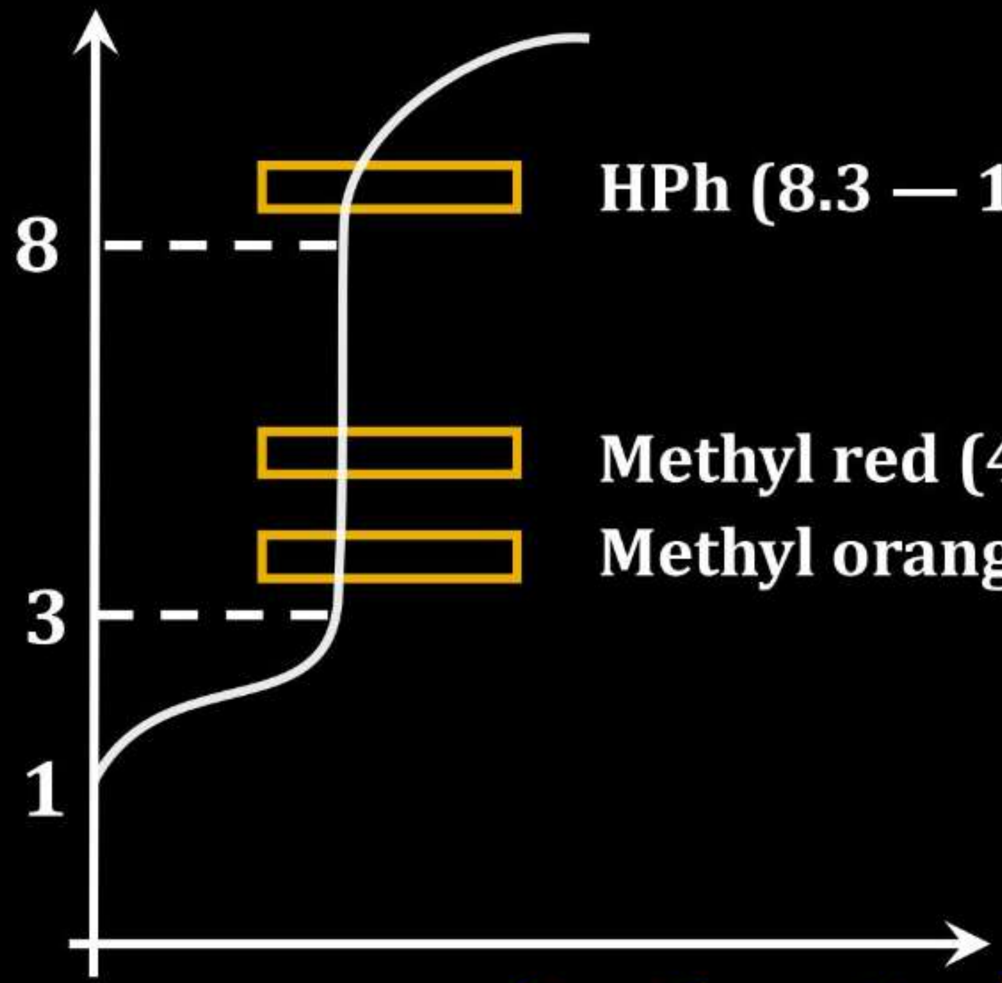
(5) Titration of Strong acid with weak Base

e.g. HCl + NH_4OH

$\boxed{3 \quad 8}$

(5) Titration of Strong acid with weak Base

e.g. $HCl + NH_4OH$



HPh (8.3 — 10)

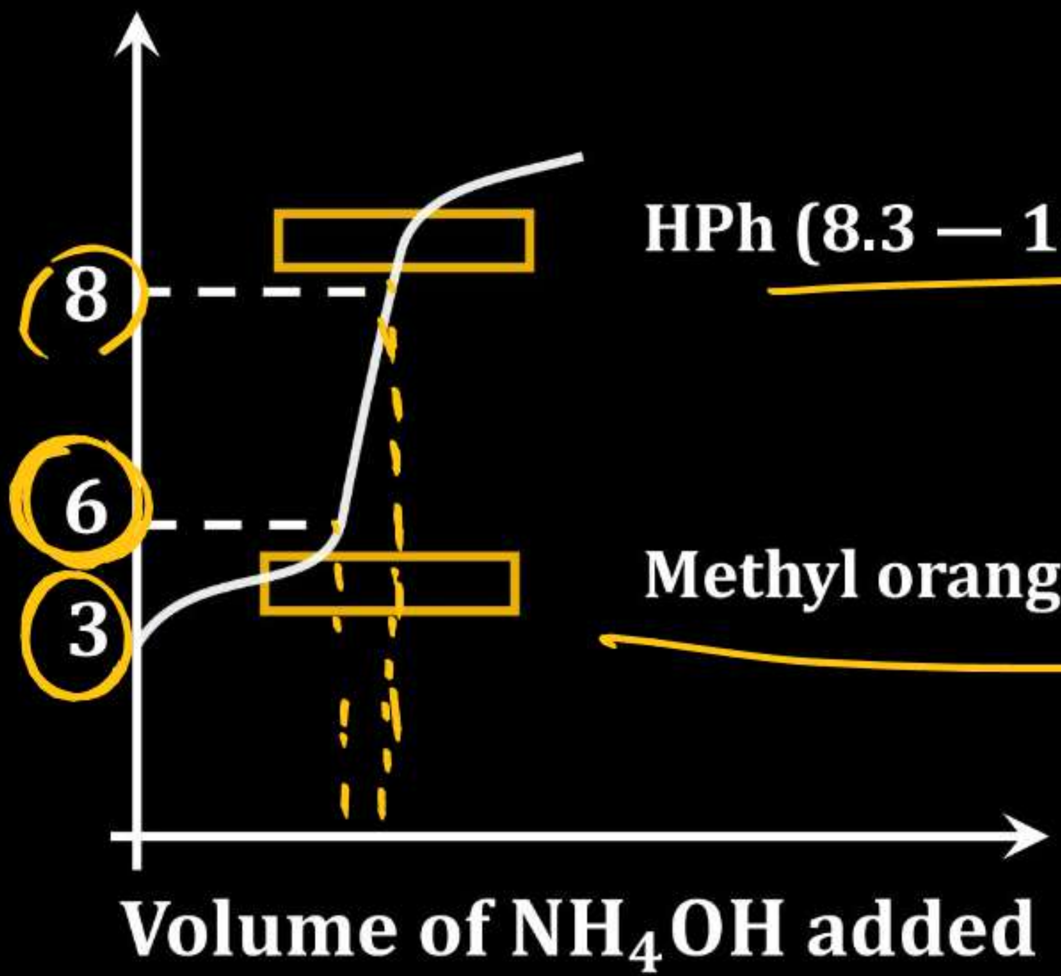
Methyl red (4.2 — 6.3) → suitable

Methyl orange (3.1 — 4.4) → suitable

Volume of ~~NaOH added~~ NH_4OH added

(6) Titration of weak acid with weak Base

e.g. $CH_3COOH + NH_4OH$



HPh (8.3 — 10) → not suitable

Methyl orange (3.1 — 4.4) → not suitable

Jump $pH_{eq} \pm 2$

Range: $pK_{In} \pm 1$

If $pK_{In} = pH_{eq}$ Condition for most suitable indicator

Double indicator acid base titration :

Example : Na_2CO_3 with HCl

$$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$$

$$NaHCO_3 + HCl \longrightarrow H_2CO_3 + NaCl$$

Double indicator acid base titration :

Example : Na_2CO_3 with HCl

$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$ pH changes from $(10-8)$

$NaHCO_3 + HCl \longrightarrow H_2CO_3 + NaCl$ pH changes from $(6-3)$

Example : Na_2CO_3 with HCl

$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$ pH changes from $(10 - 8)$

$NaHCO_3 + HCl \longrightarrow H_2CO_3 + NaCl$ pH changes from $(6 - 3)$

Let the number of moles of Na_2CO_3 = x

Find number of moles HCl required to change the colour of solution

(1) If phenolphthalein is used as indicator $n_{HCl} = x$

(2) If Methyl orange is used as indicator $n_{HCl} = 2x$

Example : Na_2CO_3 with HCl

$$\underset{x}{Na_2CO_3} + \underset{x}{HCl} \longrightarrow \underset{x}{NaHCO_3} + NaCl$$ pH changes from $(10-8)$

$$\underset{x}{NaHCO_3} + \underset{x}{HCl} \longrightarrow H_2CO_3 + NaCl$$ pH changes from $(6-3)$

Let the number of moles of $Na_2CO_3 = x$

Find number of moles HCl required to change the colour of solution

(1) If phenolphthalein is used as indicator $n_{HCl} = x = n_{Na_2CO_3}$

(2) If Methyl orange is used as indicator

$$n_{HCl} = 2x = 2 \times n_{Na_2CO_3}$$

Q. Find number of moles Na_2CO_3 if 5 moles HCl are required to change the colour of solution

(1) If phenolphthalein is used as indicator 5

(2) If Methyl orange is used as indicator 2.5 or ~~10~~

Sol. (1) If phenolphthalein is used as indicator

$$n_{Na_2CO_3} = n_{HCl} = 5$$

(2) If Methyl orange is used as indicator

$$2 \times n_{Na_2CO_3} = n_{HCl} = 5$$

2 3

Q. A mixture contains both Na_2CO_3 and $NaHCO_3$. Find number of moles of each if 2 moles HCl are required to change the colour of solution if phenolphthalein is used as indicator and 7 mole of HCl are required to change the colour of solution if Methyl orange is used as indicator separately.

Sol. (1) If phenolphthalein is used as indicator

$x = \underline{n_{Na_2CO_3}} = \underline{n_{HCl}} = 2$

$\underset{x}{Na_2CO_3} \xrightarrow{HCl} NaHCO_3$

$\underset{y}{NaHCO_3}$

(2) If Methyl orange is used as indicator

$2 \times n_{Na_2CO_3} + n_{NaHCO_3} = n_{HCl} = 7$

$n_{NaHCO_3} = 3$

$2x + y = 7$

$4 + y = 7$

Q. A mixture contains both Na_2CO_3 and $NaHCO_3$. Find number of moles of each if 2 moles HCl are required to change the colour of solution if phenolphthalein is used as indicator. Then Methyl orange is added to this solution and further titrated with HCl. 7 moles of HCl are required to change the colour of solution.

DPP-17

J-Main

Sol. (1) If phenolphthalein is used as indicator

$$n_{Na_2CO_3} = n_{HCl} = 2$$

(2) If Methyl orange is used as indicator

$$n_{Na_2CO_3} + n_{NaHCO_3} = n_{HCl} = 7$$

$$n_{NaHCO_3} = 5$$